
EE 612: OFDM For Optical Communications

Course Code	EE 612
Course Name	OFDM For Optical Communications
Approval	5th Senate (Oct 2013)

Course Outline

- **Introduction** : Mathematical Formulation -' of , an : _ OFDM Signal, Discrete FOurier Transform ImplementatiOn of OFDM, Cyclic Prefix for OFDM, Spectral Efficiency for Optical OFDM, Cross—Channel OFDMi Multiplexing without Guard Band, Complex and Real , Representations of an- OFDM Signal, Peak-to-Average Power Ratio Of OFDM Signals, Frequency Offset and Phase Noise Sensitivity, Frequency Offset Effect, Phase Noise Effect,
- **Module II:** Coding for Optical OFDM Systems Linear Block Codes, Cyclic Codes, Bose-Chaudhuri—Hocquenghem Codes, Reed— Solomon
- **Module III:** Codes, Concatenated Codes, and Product Codes, Codes on Graphs, Turbo Codes, Turbo v Product Codes, LDPC Codes, Generalized LDPC. Codes, Symbol Error rate for QAM (16, 64, 256,..., M-QAM)
- **Various Types of Optical OFDM:** Coherent Optical OFDM, Principle of CO-OFDM, Optical Transmitter Design . for CO-OFDM, Up-IDown-Conversion Design Options for CO-OFDM Systems, Optical I/Q Modulator for Linear RF-to-Optical up Conversion, Receiver
- **Module V:** Sensitivity for ,CO-OFDM, Direct Detection -Optical‘ OFDM
- **OFDM. Applications in Access Optical Network:** OFDM in Radio-over—Fiber Systems,OFDM in Passive Optical Networks,
- Wideband Signals and Optical OFD